INDEPENDENT AD7606C ENGINEERING VARIANT — ORIGINAL REV.A RETAINED

power_input

VBUS_ADC
GND
VBUS_PROT

File: power_input.kicad_sch

adc_levelshift

VIO_3V3
VA_5V
ADC_CONVST
ADC5_CONVST
ADC_RESET
ADC5_RESET
ADC_DOUTA
ADC5_DOUTA
ADC_DOUTB
ADC5_DOUTB
PWR_READY
GND
ADC_SCLK
ADC5_SCLK
ADC_CS_N
ADC5_CS_N
ADC_DOUTC
ADC5_DOUTC
ADC_DOUTD
ADC5_DOUTD
ADC_SDI
ADC5_SDI
ADC_BUSY
ADC5_BUSY
ADC_FRSTDATA
ADC5_FRSTDATA

File: adc_levelshift.kicad_sch

aux_power

PWR_GOOD
GND
VA_5V
VBUS_PROT
VDRV_12V

File: aux_power.kicad_sch

gate_inverter

GND
VBUS_BRIDGE
VDRV_12V
VIO_3V3
PWM_UH
PWM_UL
PWM_VH
PWM_VL
PWM_WH
PWM_WL
SW_W
SW_V
SW_U
PH_U
PH_V
PH_W
GATE_EN
PWR_READY
ARMED
PWR_GOOD

File: gate_inverter.kicad_sch

current_adc

GND
VA_5V
PH_U
SW_U
IU_ADC
PH_V
SW_V
IV_ADC
PH_W
SW_W
IW_ADC
VBUS_PROT
VBUS_BRIDGE
OCP_N
NTC_BOARD
NTC_BOARD_AUX
VIO_3V3

File: current_adc.kicad_sch

adc_core

IU_ADC
IV_ADC
IW_ADC
VBUS_ADC
NTC_BOARD
NTC_BOARD_AUX
GND
ADC5_CONVST
ADC5_RESET
ADC5_SCLK
ADC5_CS_N
ADC5_SDI
ADC5_DOUTA
ADC5_DOUTB
ADC5_DOUTC
ADC5_DOUTD
ADC5_BUSY
ADC5_FRSTDATA
VA_5V

File: adc_core.kicad_sch

adc_power

VA_5V
GND

File: adc_power.kicad_sch

ax7010_interface

ADC_BUSY
VIO_3V3
ADC_RESET
ADC_CS_N
ADC_DOUTA
PWM_VL
ENC_A
GATE_EN
ENC_Z
ENC_B
OCP_N
PWR_GOOD
PWM_UL
PWM_VH
PWM_UH
ADC_FRSTDATA
ADC_DOUTB
PWM_WH
GND
PWM_WL
ADC_CONVST
FAULT_CLEAR
ADC_SCLK
ADC_DOUTC
ADC_DOUTD
ADC_SDI

File: ax7010_interface.kicad_sch

ocp_latch

VIO_3V3
VA_5V
GND
OCP_N
PWR_GOOD
GATE_EN
FAULT_CLEAR
ARMED
PWR_READY

File: ocp_latch.kicad_sch

encoder

ENC_Z
VIO_3V3
GND
VA_5V
ENC_A
ENC_B

File: encoder.kicad_sch

AD7606C independent development schematic; release gates OPEN

magic-alt/fpga-servo

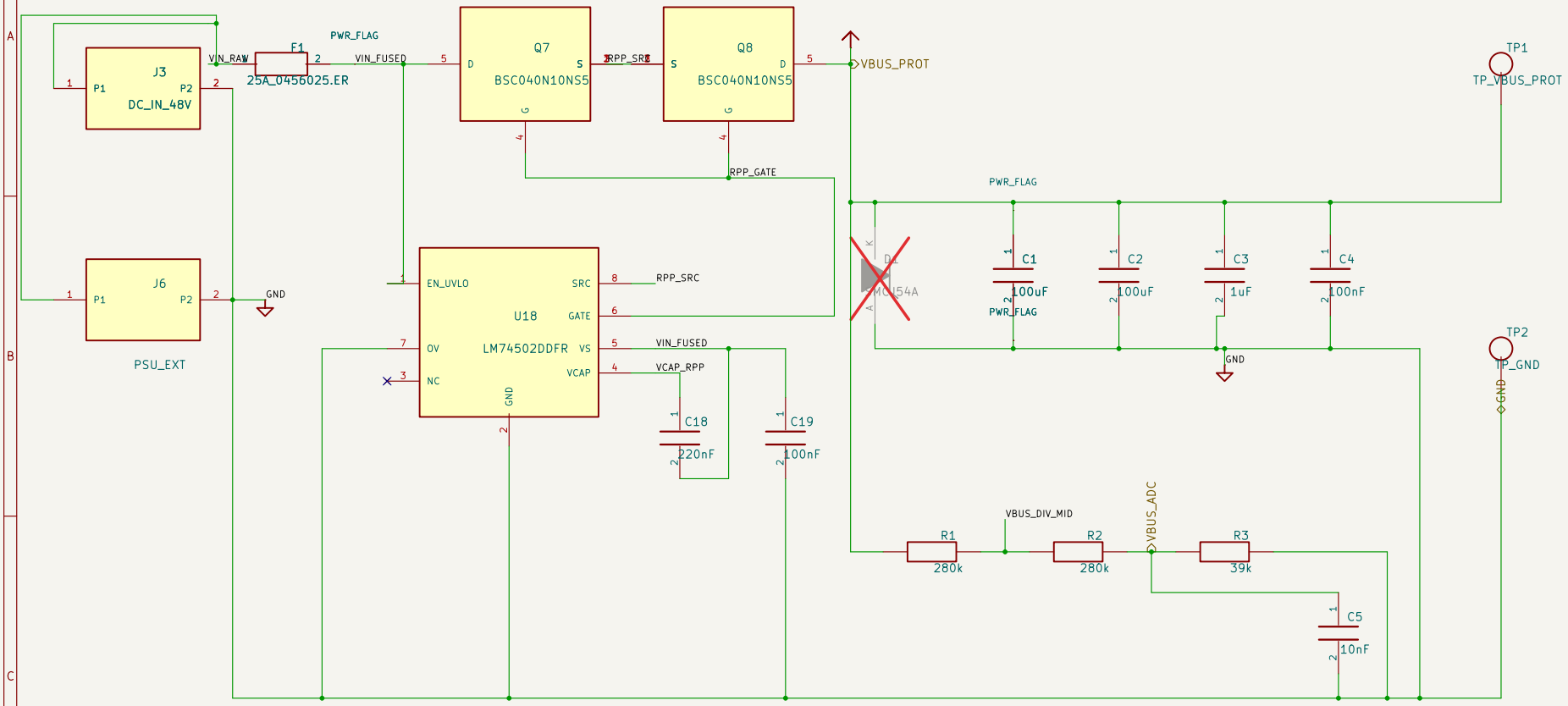
Sheet: /

File: ax7010_servo_ad7606c.kicad_sch

Title: AX7010 servo — AD7606C Rev.B

Size: A4	Date: 2026-10-17	**Rev: B-DEV**
KiCad E.D.A. 10.0.3		Id: 1/11

SOURCE + -> J3.1 -> ONBOARD F1 -> LM74502 RPP -> DC-LINK



Q7/Q8 are back-to-back N-MOSFETs driven by LM74502 for low-loss reverse-polarity protection. Reverse current is not blocked when enabled; regeneration needs a receptive source.

LM74502 OV is tied low in Rev.A1; positive surge is NOT claimed as closed. The 48V source/cable transient must be measured and kept inside the controller/downstream qualification envelope.

F1: ONBOARD 0456025.ER 25A SMD. Verify fault current, inrush and thermal coordination. D1 stays DNP pending surge qualification.

VBUS_ADC divider = 280k + 280k over 39k, plus 10nF at the ADC node. Keep the high-side resistors split for voltage stress and route the sense return away from commutation current.

Local wiring: power, gate, divider/filter and ground. See docs/schematic_full_review_2026-10-09.md.

24..48V operating bus; source transient / regeneration qualification OPEN

magic-alt/fpga-servo

Sheet: /power_input/

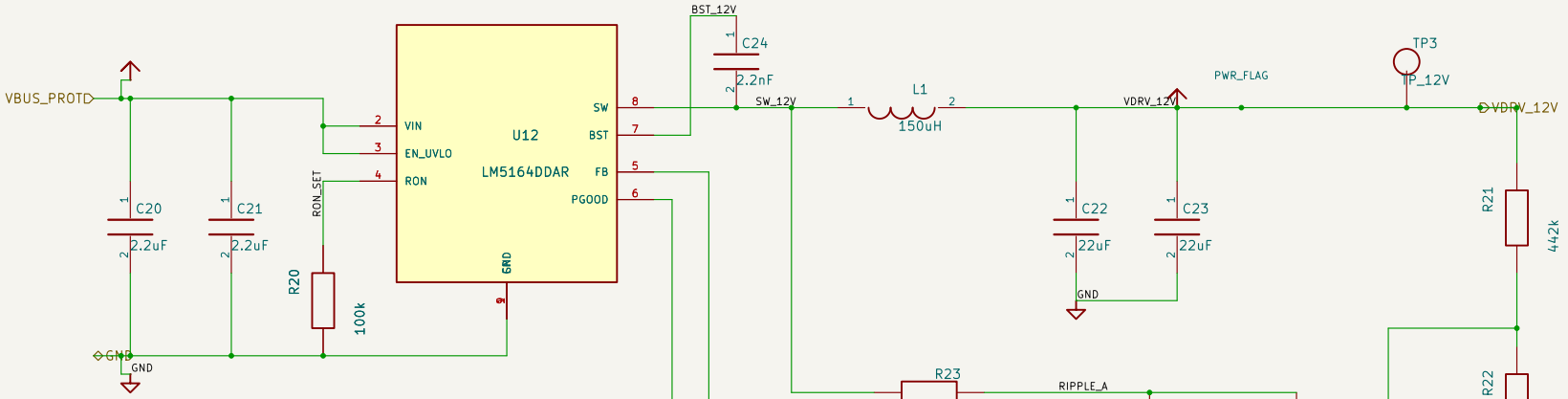
File: power_input.kicad_sch

Title: Rev.A1 — DC Input & Protection

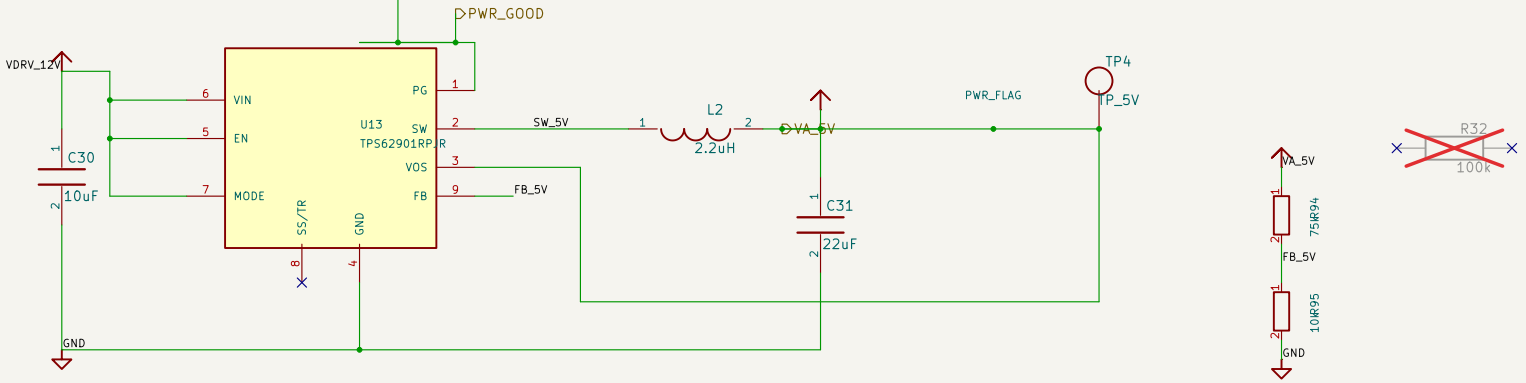
Size: A4	Date: 2026-09-29	**Rev: A1**
KiCad E.D.A. 10.0.3		Id: 2/11

VBUS_PROT -> LM5164 12V -> TPS62901 5.1V. Rail load / transient qualification OPEN.

12 V BUCK: input / bootstrap / output LC / feedback + ripple injection



5 V BUCK: precision external FB; forced PWM 2.5MHz



12V total 1A: L1=150uH; FB=442k/49.9k 0.1%; ripple=200k/3.3nF C0G. Startup / thermal qualification OPEN.

TPS62901: 5.1V nominal; FB=75k/10k; MODE=VIN: forced PWM 2.5MHz. VOS to output.

LM5164 PGOOD and TPS62901 PG share PWR_GOOD; pull up at 3.3V safety logic.

System operating input: 24..48V. Low-voltage bring-up is separately qualified.

PGOOD alone does not prove VA5 valid; U25 separately monitors VA5.

LM5164 48V->12V + TPS62901 precision 5.1V

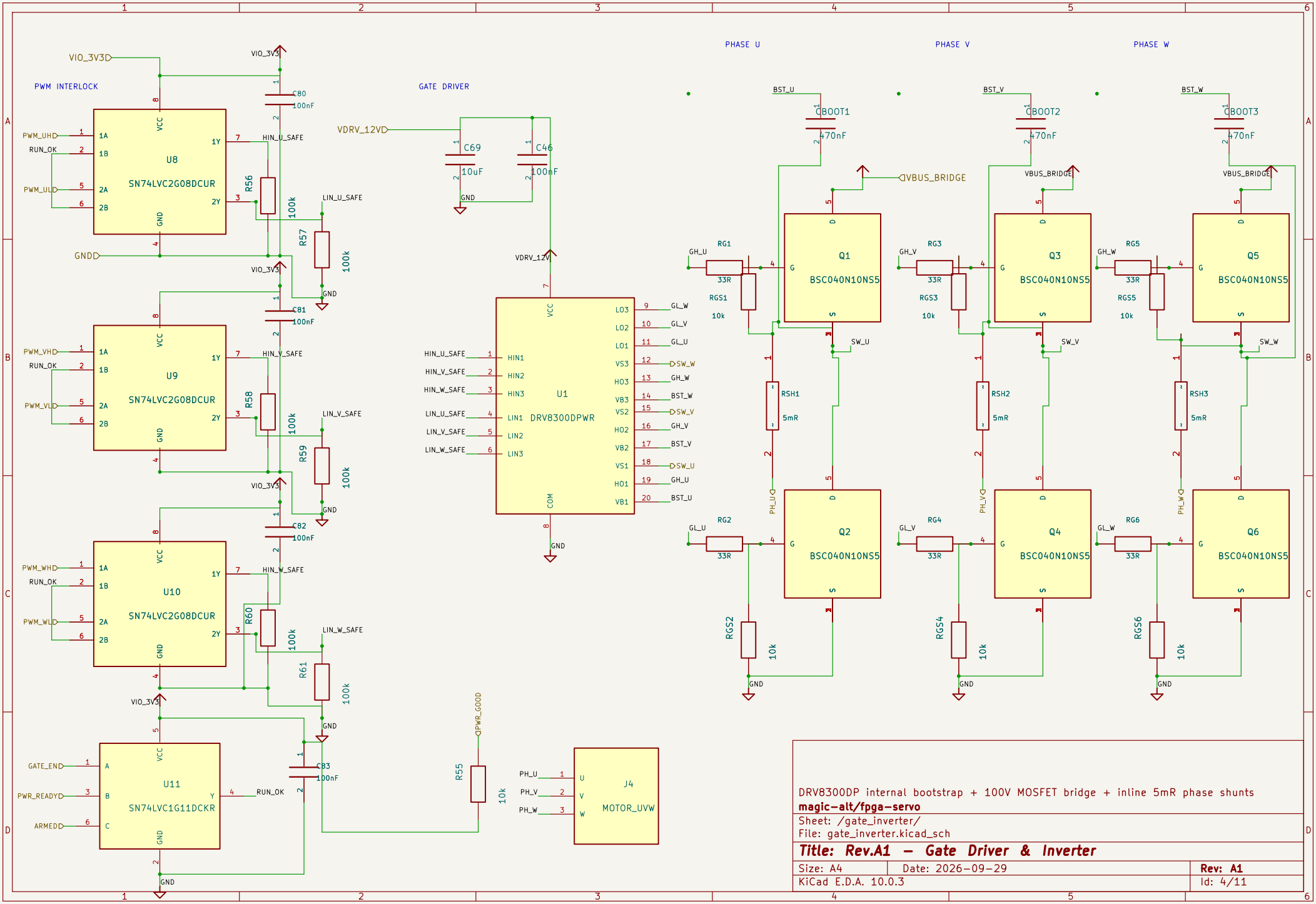
magic-alt/fpga-servo

Sheet: /aux_power/

File: aux_power.kicad_sch

Title: Rev.A1 – Auxiliary Power

Size: A4	Date: 2026-09-29	**Rev: A1**
KiCad E.D.A. 10.0.3		Id: 3/11



DRV8300DP internal bootstrap + 100V MOSFET bridge + inline 5mR phase shunts

magic-alt/fpga-servo

Sheet: /gate_inverter/

File: gate_inverter.kicad_sch

Title: Rev.A1 — Gate Driver & Inverter

Size: A4	Date: 2026-09-29	**Rev: A1**
KiCad E.D.A. 10.0.3		Id: 4/11

Phase RC: 47R / 1nF retained for first prototype; tune after PWM-noise tests

CURRENT SENSE + RC FILTER

BUS OCP: +25A NOMINAL

Bridge bus shunt after DC-link capacitors

ADC moved to dedicated core + power sheets

No reverse-current trip; latch requires explicit re-arm

AD7606C independent development schematic; release gates OPEN

magic-alt/fpga-servo

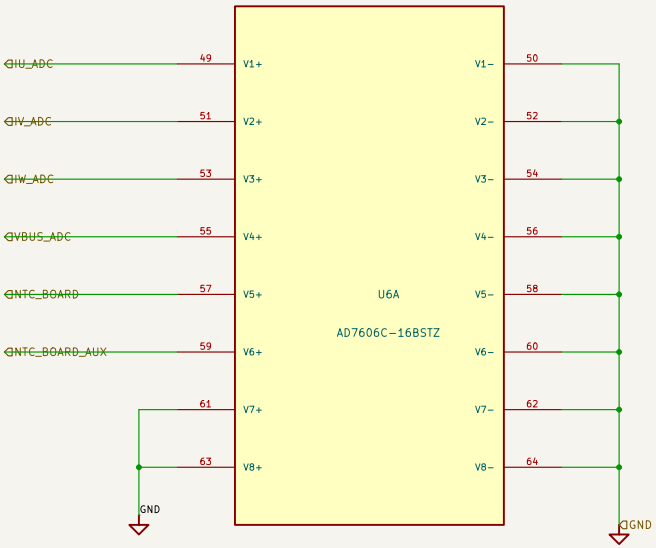
Sheet: /current_adc/

File: current_adc.kicad_sch

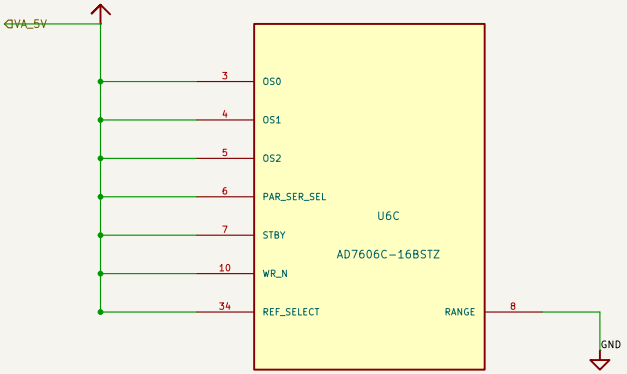
Title: Rev.B – current sensing / independent bus OCP

Size: A4	Date: 2026-10-17	**Rev: B-DEV**
KiCad E.D.A. 10.0.3		Id: 5/11

AD7606C-16 / ANALOG INPUTS



MODE STRAPS: software / serial / internal reference

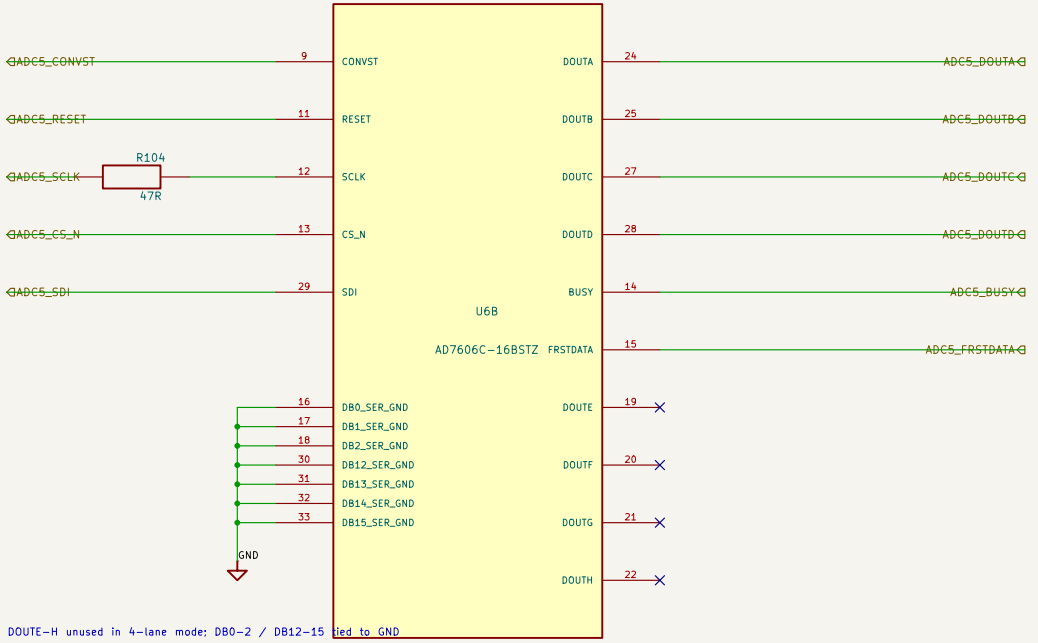


Hold GATE_EN low until reset + configuration readback pass.

Unipolar ADC codes: straight binary. See adc_profile.json.

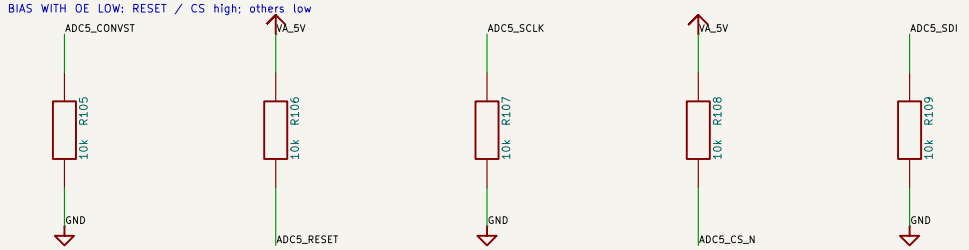
Register writes are required after power-up; straps alone do not configure ranges.

4-LANE / 5 MHz BRING-UP / 20 MHz QUALIFICATION TARGET



DOUTE-H unused in 4-lane mode; DB0-2 / DB12-15 tied to GND

BIAS WITH OE LOW: RESET / CS high; others low



AD7606C independent development schematic; release gates OPEN

magic-alt/fpga-servo

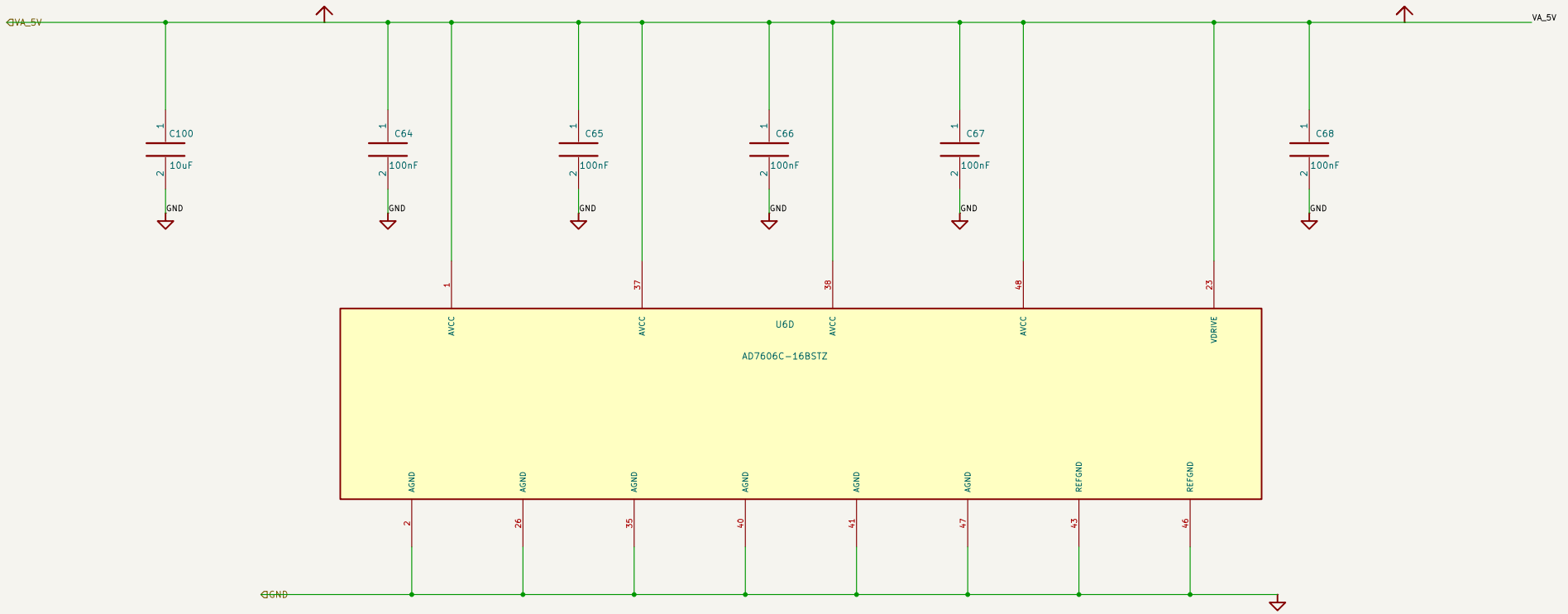
Sheet: /adc_core/

File: adc_core.kicad_sch

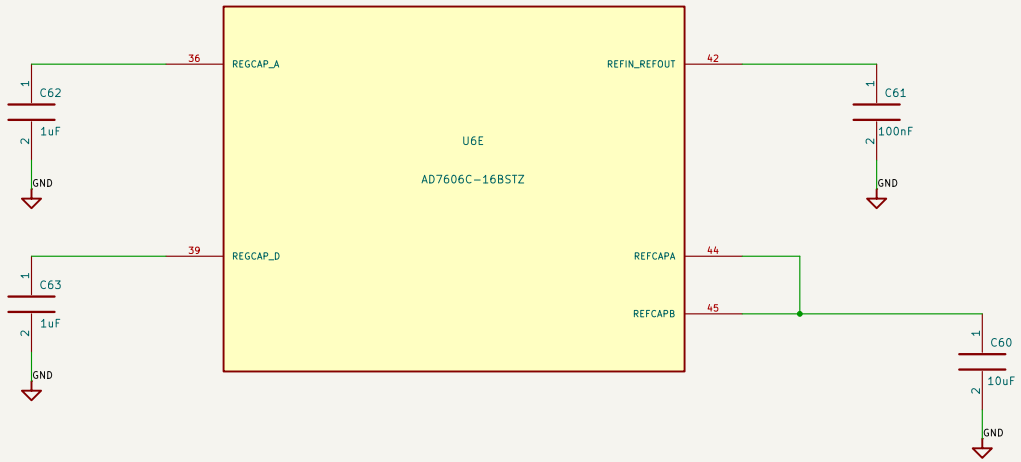
Title: Rev.B – AD7606C input / serial / startup

Size: A4	Date: 2026-10-17	**Rev: B-DEV**
KiCad E.D.A. 10.0.3		Id: 6/11

ADC SUPPLIES: 100 nF AT EACH AVCC + 10 uF LOCAL BULK



REGCAP A/D SEPARATE; REFCAPA/B SHORTED; NO EXTERNAL LOAD ON THESE NODES



AD7606C independent development schematic; release gates OPEN

magic-alt/fpga-servo

Sheet: /adc_power/

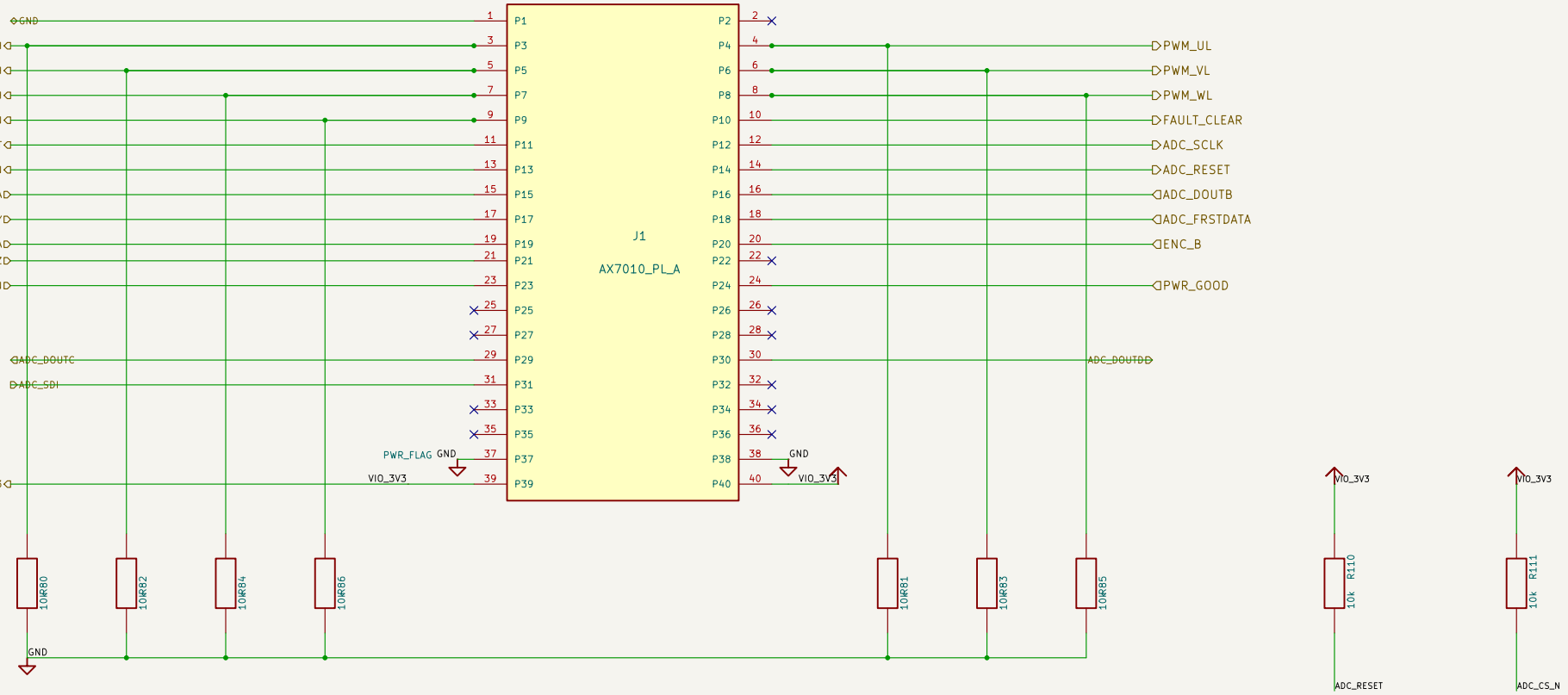
File: adc_power.kicad_sch

Title: Rev.B — AD7606C supply / reference / LDO bypass

Size: A4	Date: 2026-10-17	**Rev: B-DEV**
KiCad E.D.A. 10.0.3		Id: 7/11

ALINX AX7010 (2022): active J10 interface + physical reset-state pull-down branches

J1 = AX7010 J10



R80..R86: each PWM and GATE_EN line has its own 10k pull-down.

J10 pin 2 (+5 V): NC. J2 removed; only J1 interfaces to AX7010.

AD7606C: J10.29=DOUTC (U12), .30=DOUTD (T12), .31=SDI (T10)

R110/R111: hold RESET/CS high at TXU inputs while FPGA is Hi-Z.

AD7606C independent development schematic; release gates OPEN

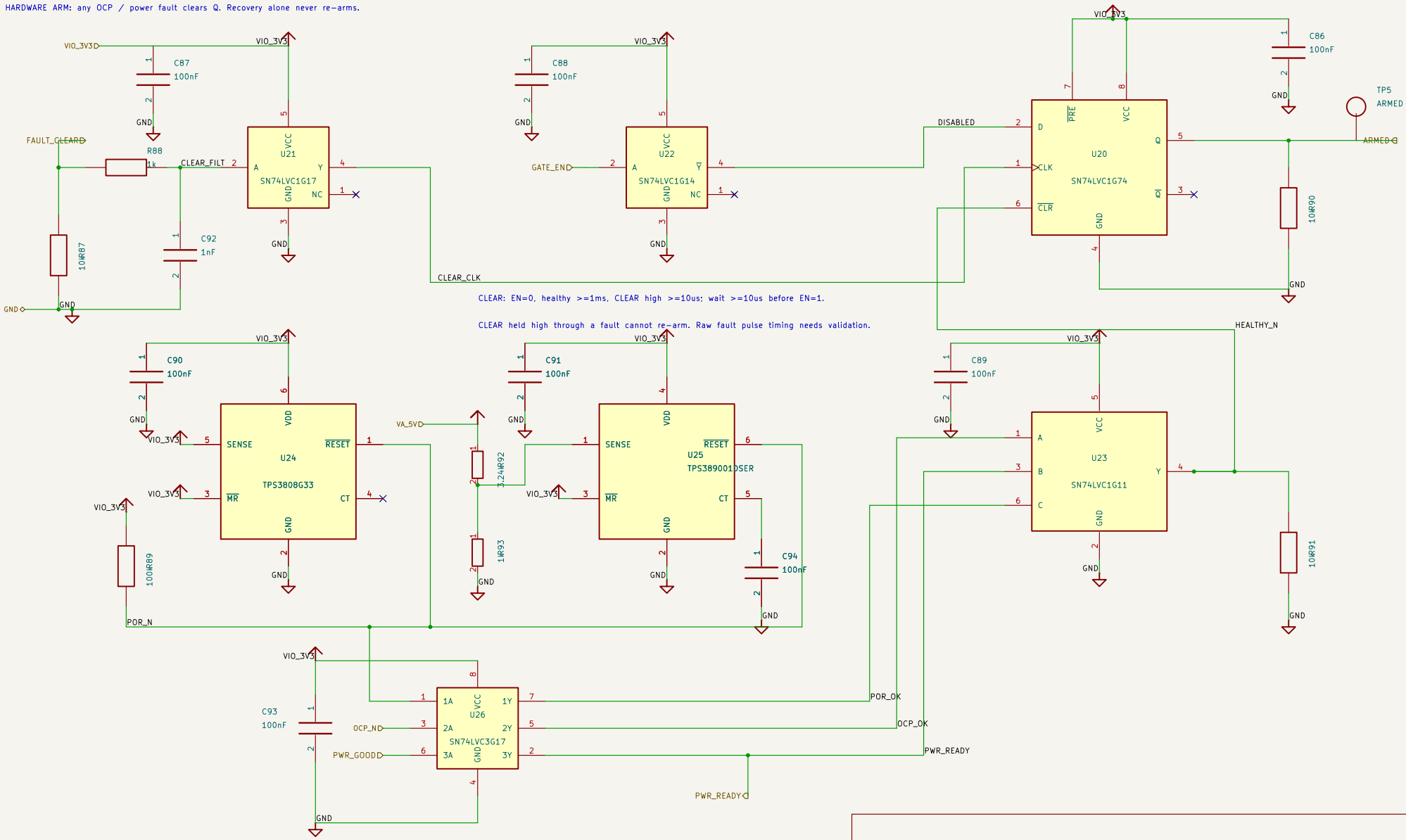
magic-alt/fpga-servo

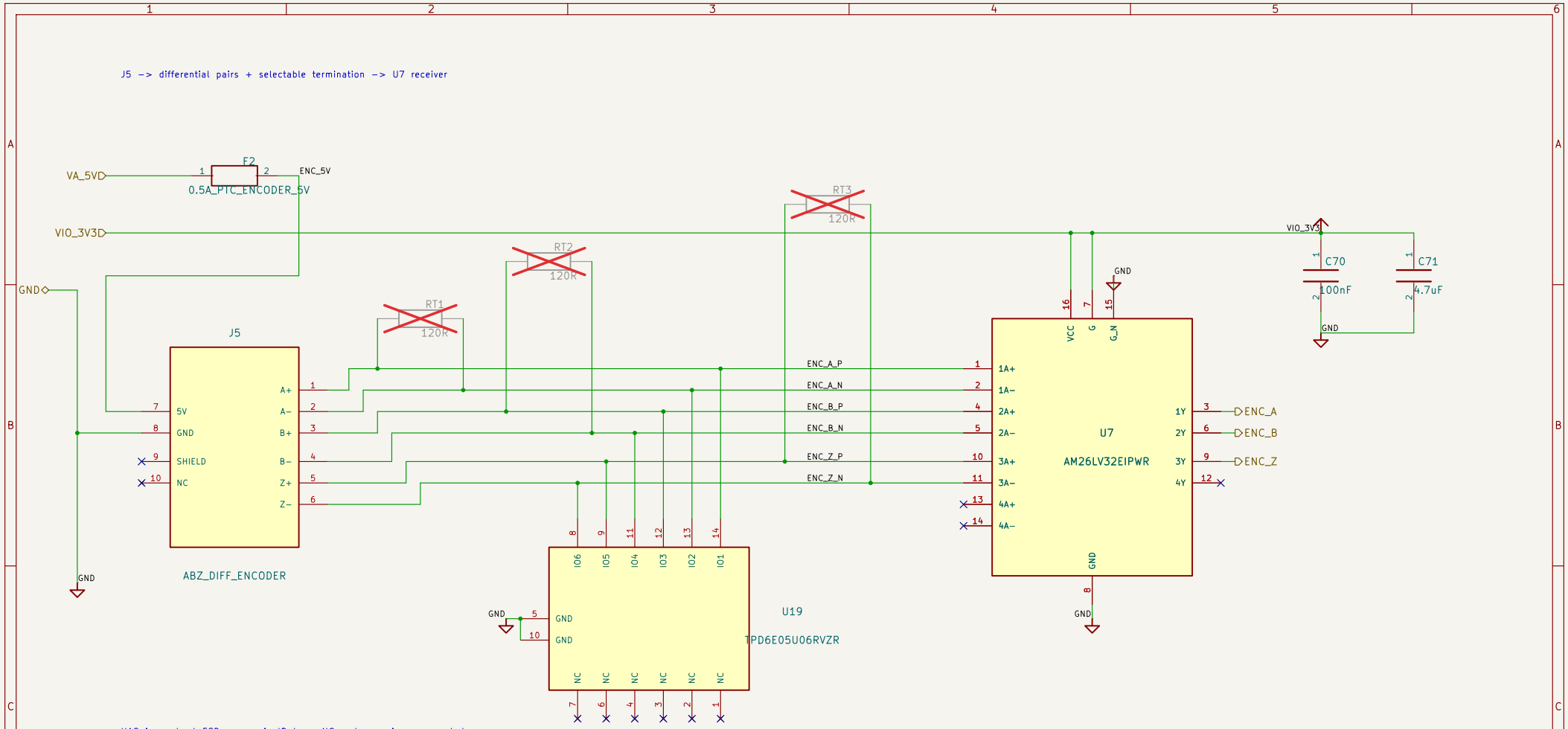
Sheet: /ax7010_interface/

File: ax7010_interface.kicad_sch

Title: Rev.B – AX7010 2022 J10 / four-lane ADC

Size: A4	Date: 2026-10-17	**Rev: B-DEV**
KiCad E.D.A. 10.0.3		Id: 8/11


HARDWARE ARM: any OCP / power fault clears Q. Recovery alone never re-arms.
VIO_3V3
C87
100nF
GND
FAULT_CLEAR
R88
1k
CLEAR_FILT
U21
SN74LVC1G17
VCC
A
Y
GND
NC
C92
1nF
10kR87
C88
100nF
U22
SN74LVC1G14
GATE_EN
C86
100nF
TP5
ARMED
DISABLED
U20
SN74LVC1G74
D
CLK
CLR
PRE
Q
10kR90
CLEAR_CLK
CLEAR: EN=0, healthy >=1ms, CLEAR high >=10us; wait >=10us before EN=1.
CLEAR held high through a fault cannot re-arm. Raw fault pulse timing needs validation.
HEALTHY_N
C90
100nF
U24
TPS3808G33
SENSE
RESET
MR
CT
VDD
VA_5V
3.24kR92
1kR93
C91
100nF
U25
TPS3890010SER
C89
100nF
U23
SN74LVC1G11
A
B
C
Y
10kR91
100kR89
POR_N
C94
100nF
U26
SN74LVC3G17
1A
2A
3A
1Y
2Y
3Y
C93
100nF
OCP_ND
PWR_GOODD
POR_OK
OCP_OK
PWR_READY
U24 CT open: 12..28ms. U25 CT=100nF: ~107ms; VA5 trip nominal 4.876V. Dynamic qualification OPEN.
U20..U26: local 100nF bypass; reset release passes through U26 Schmitt buffers.
magic-alt/fpga-servo
Sheet: /ocp_latch/
File: ocp_latch.kicad_sch
Title: Hardware OCP latch / supervised re-arm
Size: A4
Date: 2026-10-09
Rev: A1
KiCad E.D.A. 10.0.3
Id: 9/11



A/B/Z are differential RS-422 pairs. 120R termination is fitted only when this board is the cable end.

U19: RVZ-14 / Texas_R-PUSON-N14. IO=14,13,12,11,9,8; GND=5,10; NC=1,2,3,4,6,7. Place beside J5.

AM26LV32E is powered from FPGA VIO_3V3 so its outputs cannot over-drive an unpowered AX7010 domain.

Local wiring: ABZ pairs, termination, ESD, receiver and encoder power.

RS-422 receiver + selectable termination + interface protection

magic-alt/fpga-servo

Sheet: /encoder/

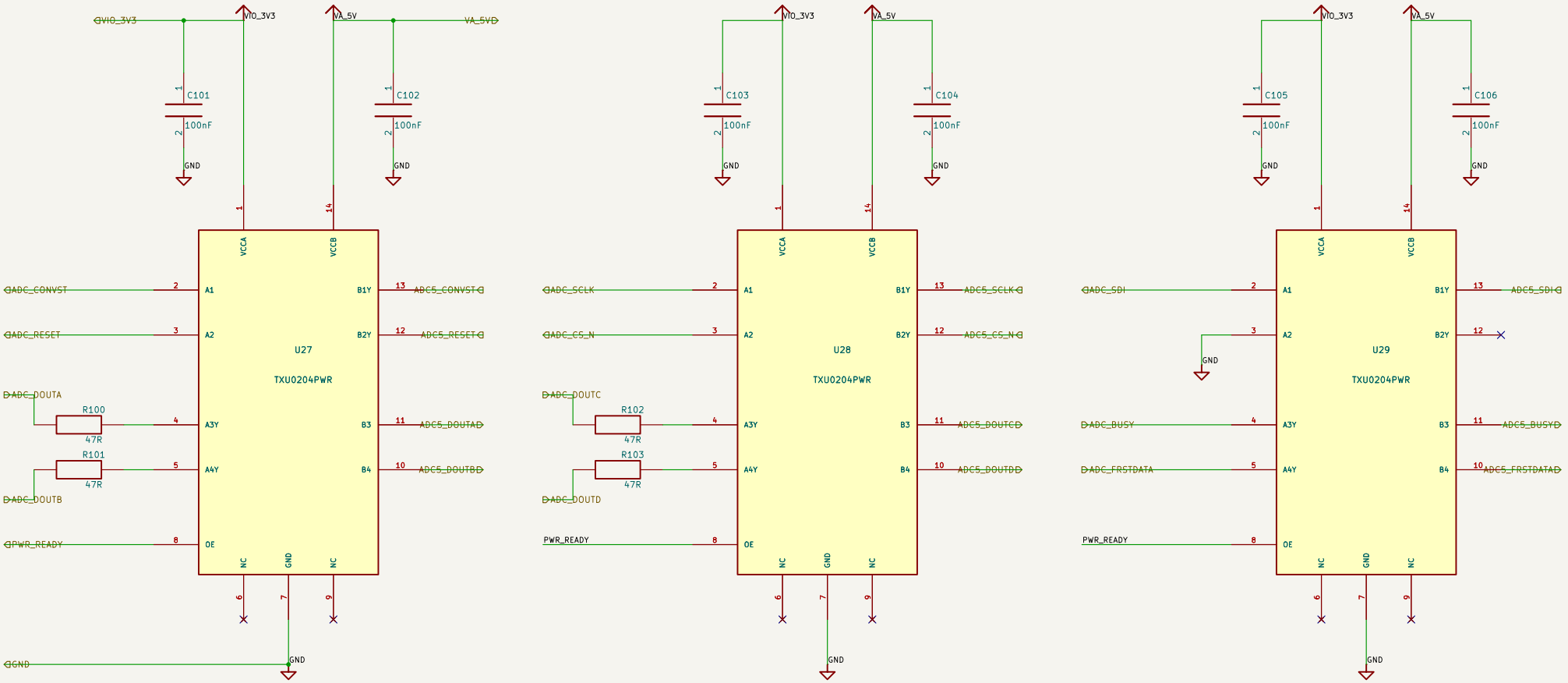
File: encoder.kicad_sch

Title: Rev.A1 — Differential ABZ Encoder

Size: A4	Date: 2026-09-29	**Rev: A1**
KiCad E.D.A. 10.0.3		Id: 10/11

ADC POWER-DOMAIN ISOLATION: FPGA 3.3 V <-> ADC VA_5V

U6 AVCC = VDRIVE = VA_5V; never connect AX7010 3.3 V directly to U6.23



OE follows PWR_READY: translators disable on supervised rail faults.

TI Ioff / VCC disconnect prevents back-power when either supply is off.

RESET / CS pulls are on the ADC side. Firmware must apply full RESET after OE is enabled.

Translate delay, J10 flight time and ADC delay must all be included in Vivado I/O timing.

AD7606C independent development schematic; release gates OPEN

magic-alt/fpga-servo

Sheet: /adc_levelshift/

File: adc_levelshift.kicad_sch

Title: Rev.B – ADC voltage / partial-power isolation

Size: A4	Date: 2026-10-17	**Rev: B-DEV**
KiCad E.D.A. 10.0.3		Id: 11/11